

MEMO



ST BENEDICTS COLLEGE
NOVEMBER EXAMINATIONS

ADVANCED PROGRAMME MATHEMATICS GR 11

Examiner: Mrs D Park-Dobell

Time: 1 hrs

Moderator: Mrs M Povall,

Total: 100

QUESTION NO	DESCRIPTION	MAXIMUM MARK	MARK OBTAINED
1	VENN DIAGRAMS	19	
2	SCATTER PLOTS	10	
3	COUNTING METHODS	25	
4	PROBABILITY AND COUNTING METHODS	13	
5	NORMAL DISTRIBUTION CURVE	18	
6	NORMAL DISTRIBUTION CURVE	15	
	TOTAL	100	

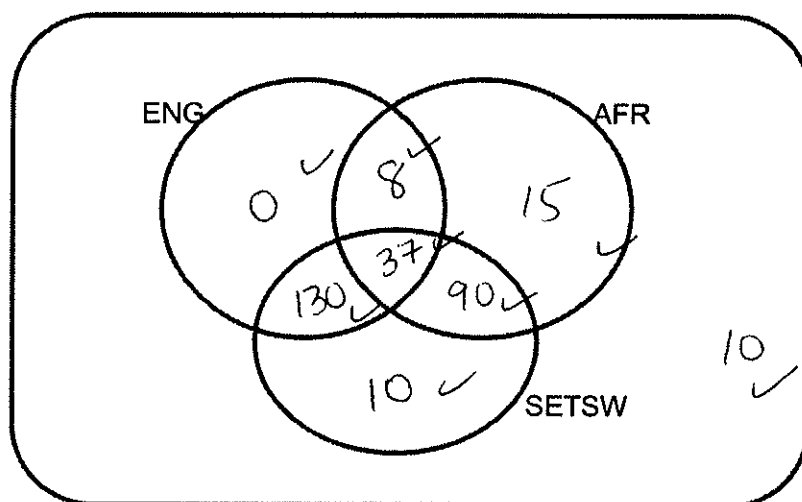
INSTRUCTIONS TO CANDIDATES

1. Round all answers off to 3 decimals where applicable.
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QUESTION 1

There are 300 Grade 10 pupils at a school at Soshanguve High School. Of these pupils, 267 take Setswana, 175 take English and 150 take Afrikaans. Also, 37 pupils take all three languages, 167 take English and Setswana, 127 take Setswana and Afrikaans and no pupil takes only English. Draw a Venn diagram to illustrate this data and calculate the probability that a pupil, chosen at random takes:

(8)



- a) All 3 languages.

(2)

$$\frac{37}{300} \quad \checkmark$$

- b) Setswana and English but not Afrikaans

(3)

$$\frac{130}{300} \checkmark = \frac{13}{30} \checkmark$$

- c) None of the languages listed above

(3)

$$\frac{10}{300} \checkmark = \frac{1}{30} \checkmark$$

- d) Only Afrikaans.

(3)

$$\frac{15}{300} \checkmark = \frac{1}{20} \checkmark$$

QUESTION 2

The heights, x , and weights, y , of ten Grade 12 boys were measured and found to be:

HEIGHTS CM	160	161	180	183	189	188	185	179	177	183
WEIGHTS KG	75	80	81	85	81	86	90	87	88	90

- a) Determine an equation for the regression line for this scatter plot: (4)

$$y = 0,3049x + 29,883$$

- b) Calculate the correlation coefficient and comment on its value. (3)

$$r = 0,6346 \quad \checkmark \quad \text{moderate} \quad \checkmark \\ \text{positive} \quad \checkmark$$

- c) Predict the height of a Grade 12 learner who weighs 84kg. (3)

$$\checkmark \checkmark \checkmark \\ \text{height} = 177,516 \text{ cm}$$

QUESTION 3

- a) Given that $P(A|B) = \frac{2}{3}$. If A and B are Independent events and $P(A \cap B) = \frac{1}{3}$, determine:

i) $P(B)$ $P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \checkmark$ (5)

$$\checkmark \quad \frac{2}{3} \times P(B) = \frac{1}{3} \quad \checkmark$$

$$\therefore P(B) = \frac{1}{3} \times \frac{3}{2} = \frac{1}{2} \quad \checkmark \quad \checkmark$$

ii) $P(A)$ (6)

$$P(A) \cdot P(B) = P(A \cap B) \quad \checkmark \checkmark$$

$$\frac{1}{2} P(A) = \frac{1}{3} \quad \checkmark \checkmark$$

$$P(A) = \frac{1}{3} \times \frac{2}{1} = \frac{2}{3} \quad \checkmark \checkmark$$

- b) In how many ways can a committee of 5 St Benedicts learners and 3 St John's learners be chosen from a total of 11 St Benedict's and 8 St John's boys (6)

$${}^{11}C_5 \times {}^8C_3 = 25872$$

- c) At this year's matric dance, 12 people (6 couples) are arranged at random in a line for a photograph.

- i) In how many ways can these people be arranged? (2)

$$12!$$

- ii) Determine the probability that each learner is standing next to his/her partner. (6)

$$\frac{6! \times 2^6}{12!} = \frac{46080}{12!}$$

$$= 0,0000962$$

QUESTION 4

- a) The chess team have been given a probability of 0,7 in winning a tournament. If they have 4 tournaments during the year, what is the probability that they will win 3 of these. (7)

$${}^4C_3 \times 0,7^3 \times 0,3^1 = 0,4116$$

- b) Nine St Benedict's boys are eager to be selected for a social tennis doubles team consisting of four boys. Two of the boys refuse to play together but are happy to be paired with any of the other boys. In how many ways can the selection of these boys be made? (6)

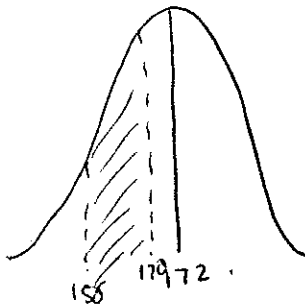
$$9C4 - 7C2 = 105$$

$$7C_4 + \binom{2}{1}\binom{7}{3} = 105$$

QUESTION 5

In a certain 2013 Grade 11 group, the height of the learners follow a normal distribution with a mean of 172cm and a variance of 25cm^2 .

- a) What percentage of the boys will have a height between 156cm and 170cm. (10)



$$\sigma = 5 \checkmark$$

$$z = \frac{156 - 172}{5} \checkmark$$

$$= -3,2 \checkmark$$

$$P(z > -3,2) = 0,49931 \checkmark$$

$$z = -\frac{2}{5} \checkmark$$

$$= -0,4 \checkmark$$

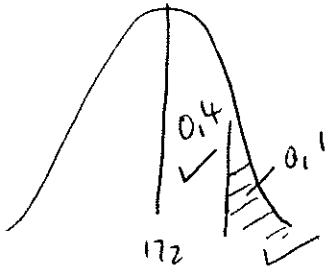
$$P(z > -0,4) = 0,1554 \checkmark$$

$$\therefore 0,49931 - 0,1554 \checkmark$$

$$= 0,34391 \checkmark$$

$$= 34,391\% \checkmark$$

- b) If only the tallest 10% of the boys will be chosen for a certain task, what is the minimum height that will be allowed for the task. (8)



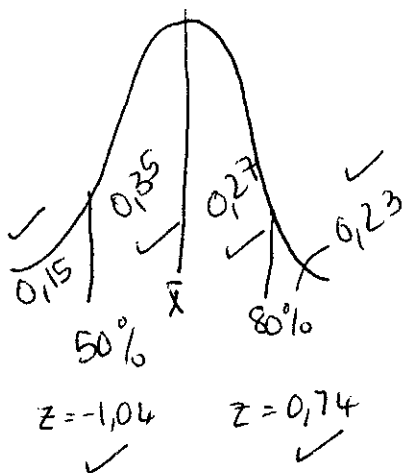
$$P(P=0.1) = 1,28 \quad \checkmark \checkmark$$

$$1,28 = \frac{x - 172}{5} \quad \checkmark \checkmark$$

$$x = 178,4 \quad \checkmark$$

QUESTION 6

In a final APM Paper, 23% of the learners attained a mark of above 80%, and 15% of the learners got results lower than 50%. Determine the mean and standard deviation for this examination. (15)



$$-1,04 = \frac{50 - \bar{x}}{\sigma} \quad \checkmark \checkmark$$

$$0,74 = \frac{80 - \bar{x}}{\sigma} \quad \checkmark \checkmark$$

$$-1,04\sigma = 50 - \bar{x} \quad \checkmark \dots (1)$$

$$0,74\sigma = 80 - \bar{x} \quad \checkmark \dots (2)$$

$$(1) = (2)$$

$$-1,78\sigma = -30$$

$$\sigma = 16,85 \quad \checkmark \checkmark$$

$$\therefore \bar{x} = 67,53\% \quad \checkmark$$